SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I Kadiri sankar narayana (192325085) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

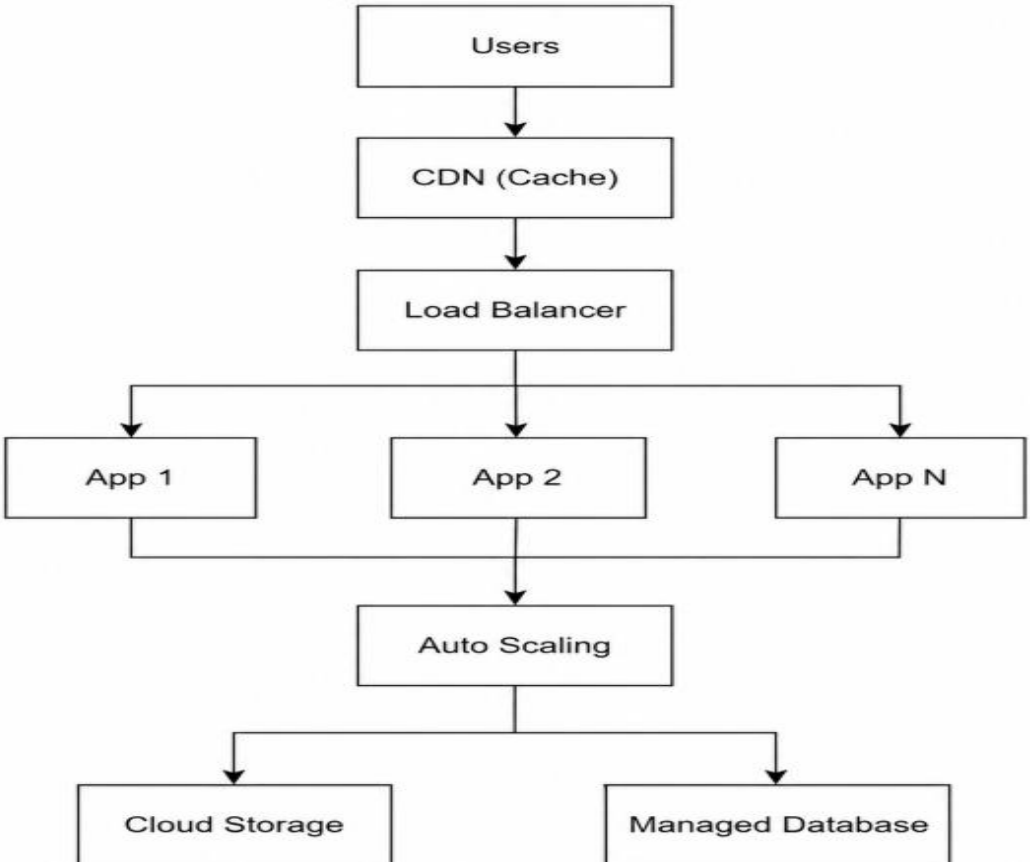
Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1)

A media company can move its video streaming platform to the cloud by using multiple cloud servers instead of depending on a single server. The videos can be stored in cloud storage, and a Content Delivery Network (CDN) can be used to keep copies of videos in different locations. This helps users watch videos from the nearest server, reducing buffering and improving streaming speed.

To manage millions of users, a load balancer is placed in front of the application servers. It distributes user requests evenly across all available servers, so no single server becomes overloaded. This improves performance and keeps the service available even during peak traffic.

Auto-scaling automatically adds more servers when the number of users increases and removes extra servers when the traffic decreases. This ensures smooth video streaming during busy hours while avoiding unnecessary resource usage when demand is low.



For cost optimization, the company can:

- Use auto-scaling to pay only for the resources that are needed.
- Store videos in low-cost cloud storage.
- Use a CDN to reduce bandwidth usage and improve delivery speed.
- Monitor resource usage regularly and remove unused resources.

- Use reserved or spot instances for workloads that are predictable or non-critical to reduce cloud costs.

2)

A fintech startup can use Kubernetes as a container orchestration platform to manage its applications across multiple cloud providers. Kubernetes helps deploy, monitor, and manage containers in the same way on different clouds, making the application consistent and easier to maintain.

Service discovery can be handled by Kubernetes using its built-in DNS and Services. This allows containers to find and communicate with each other automatically without using fixed IP addresses.

For scaling, Kubernetes supports automatic scaling. When the number of users increases, it creates more container instances to handle the traffic. When the demand decreases, it removes the extra containers. This ensures good performance and efficient use of resources.

The benefits of a multi-cloud deployment are:

- Avoids dependency on a single cloud provider.
- Improves availability and reliability.

- Better disaster recovery if one cloud service fails.
- Flexibility to choose the best services from different providers.
- Can improve performance by deploying applications closer to users.

The risks of a multi-cloud deployment are:

- Managing multiple cloud platforms is more complex.
- Security policies must be maintained across all providers.
- Monitoring and troubleshooting become more difficult.
- Data transfer between cloud providers may increase costs.
- Requires skilled professionals to manage the environment.

3)

If an e-learning platform loses data because of accidental deletion, the best solution is to use cloud backup, replication, and versioning. The platform should take regular backups and store them safely in the cloud. Data replication keeps another copy of the data on a different server or region, so even if one copy is lost, the other one is still available. Versioning is also useful because it saves older versions of files, making it easy to restore deleted or changed data.

To achieve Recovery Time Objective (RTO), the backup data should be ready for quick recovery. Since the data is already replicated, the system can switch to the backup copy and continue working with very little downtime.

To achieve Recovery Point Objective (RPO), backups should be taken frequently, and data should be replicated continuously. This helps reduce the amount of data lost before the failure happens.



To maintain business continuity, the organization should:

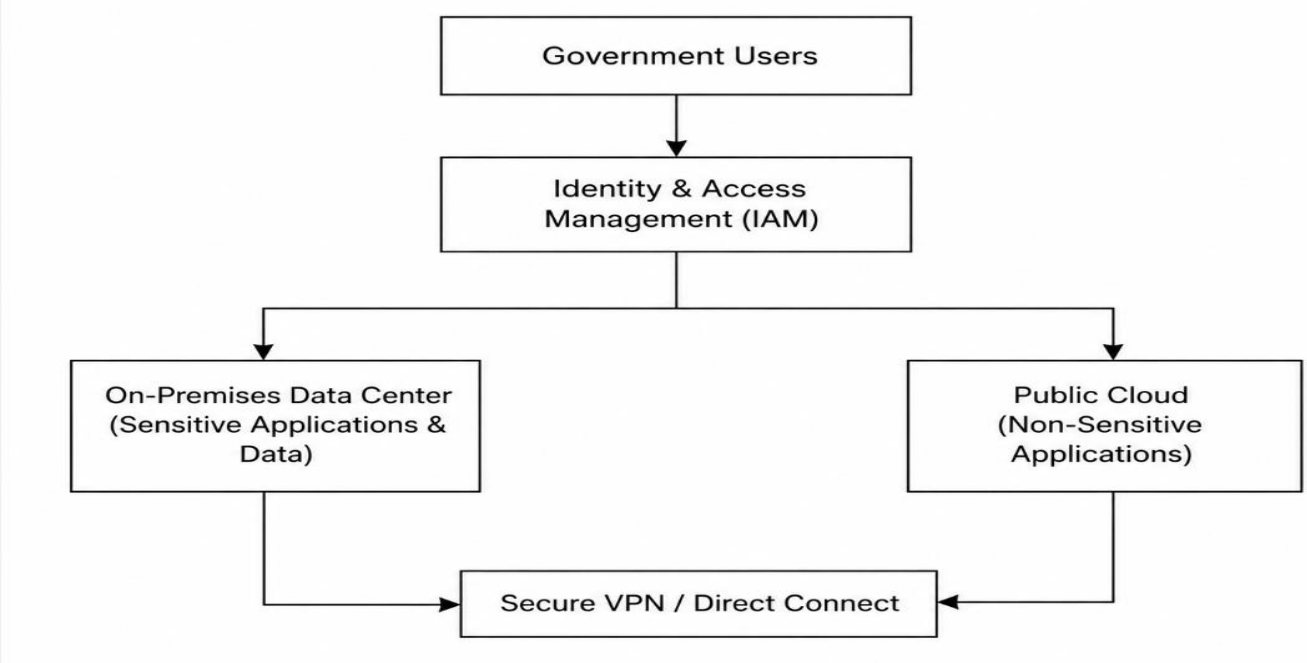
- Take automatic backups regularly.
- Store backup copies in different locations.
- Enable versioning to recover deleted files.
- Test the recovery process from time to time.
- Monitor the system and keep a disaster recovery plan ready.

4)

A government agency can use a hybrid cloud by keeping sensitive data and important applications on its own on-premises servers, while storing non-sensitive data and public services in the cloud. Both environments can be connected through a secure VPN or a private network so that data can be shared safely between them.

For identity management, the agency can use a single login system for all employees. Role-based access control should be used so that each employee can access only the information required for their work. Multi-factor authentication can also be added to improve security.

Encryption should be used to protect the data. Data should be encrypted while it is stored and also while it is being transferred between the cloud and the on-premises servers. This helps prevent unauthorized users from reading sensitive information.



Managing a hybrid cloud has some challenges. It is difficult to manage both cloud and on-premises systems together. Data synchronization, maintaining security policies, and meeting government compliance requirements require continuous monitoring and proper planning.

By using a secure connection, strong identity management, and encryption, the government agency can safely manage both environments. This provides better security for sensitive data while also taking advantage of the flexibility and scalability of cloud services.

5)

(a) Propose a multi-region deployment strategy in the cloud.

Deployment Strategy

- Deploy the application in multiple cloud regions to eliminate a single point of failure.
- Use a Global Load Balancer to route user requests to the nearest healthy region.
- Deploy application servers and databases in each region.
- Configure Active-Active or Active-Passive deployment for continuous service availability.
- Ensure each region has sufficient resources to handle user traffic if another region fails.

(b) Explain how data replication and failover mechanisms should be configured.

Data Replication

- Enable real-time database replication between all cloud regions.
- Replicate application data and storage continuously to keep data synchronized.
- Maintain backup copies in secondary regions for disaster recovery.

Failover Mechanism

- Configure automatic failover to redirect traffic if the primary region becomes unavailable.
- Use health checks to detect failures in servers or regions.
- Automatically switch users to the healthy region with minimal downtime.
- Regularly test failover procedures to ensure reliable disaster recovery.

(c) Discuss monitoring and alerting strategies for proactive issue detection.

Monitoring

- Monitor CPU, memory, storage, and network utilization.
- Track application response time and server health.
- Monitor database replication status and backup operations.
- Use centralized logging to analyze application and infrastructure events.

Alerting

- Configure alerts for server failures, high CPU usage, network latency, and storage issues.
- Send notifications through email, SMS, or messaging platforms to administrators.
- Generate alerts for replication failures, backup failures, and security incidents.
- Use dashboards to continuously monitor the health of all cloud regions.